

Week 2: Data Basics

Day 1: Understanding Data & Datasets

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8th Oct 2026

Abstract

Welcome to Day 1 of Week 2! Today, we lay the groundwork for Artificial Intelligence by understanding its most critical component: Data. This session is designed for non-programmers to grasp how data is structured, categorized, and stored before it can be used to train Machine Learning models.

- **Core Concepts:** Defining data and understanding its role as the "fuel" for AI.
- **Data Classification:** Differentiating between structured and unstructured data formats.
- **Data Types:** Identifying text, numerical, categorical, image, and audio data.
- **Datasets & CSVs:** Understanding the anatomy of tabular data and Comma-Separated Values files.
- **Practical Setup:** Navigating datasets using Microsoft Excel and Google Sheets.

Session Objectives

By the end of today's session, learners will be able to:

- Define what data is and explain why it is the foundation of AI and ML.
- Differentiate between structured and unstructured data with real-world examples.
- Identify and categorize various data types (text, numbers, images, audio).
- Understand the structure and purpose of CSV files in the AI pipeline.
- Identify the anatomy of a dataset (rows, columns, and headers).
- Open, navigate, and inspect a raw dataset using spreadsheet tools.

Part 1: Introduction to Data in AI

"The Foundation of AI – What are we working with?"

What is Data?

- **Definition:** Data consists of facts, statistics, measurements, or observations collected for reference, analysis, or decision-making.
- In the digital world, data is any information that can be digitized and processed by a computer.
- Data is the raw material from which Artificial Intelligence and Machine Learning models learn patterns.
- Without data, an AI model is like an engine without fuel—it simply cannot function or make predictions.

Key Takeaway

AI does not inherently "know" things; it "learns" from examples. Data provides those examples.

Data: The Fuel of AI & ML

- The quality, quantity, and relevance of data directly dictate the intelligence and accuracy of an AI model.
- Data tells the model what patterns to look for (e.g., features of a spam email, indicators of customer churn, or signs of crop disease).
- **The Mantra:** "Data is the new oil." However, just like crude oil, raw data must be refined and processed to be truly useful.

The Reality of AI Projects

Data collection and preparation often take up to 80% of a Data Scientist's time. The actual model training is just the final step in the pipeline.

The AI Learning Process

From Raw Information to Intelligence

1. **Data Collection:** Gathering raw facts (e.g., sales receipts, customer feedback).
2. **Data Processing:** Organizing and cleaning the information so a computer can read it.
3. **Model Training:** Feeding the processed data into an ML algorithm to find patterns.
4. **Prediction/Action:** The AI uses the learned patterns to make decisions on new, unseen data.

Today, we focus entirely on Step 1 and Step 2: Understanding the raw material and how it is structured.

Structured vs. Unstructured Data

Not all data is created equal. In AI, we broadly classify data into two main categories:

Structured Data

- Highly organized.
- Formatted in a predefined, tabular manner.
- Easily searchable using traditional tools.

Unstructured Data

- Free-form and messy.
- Lacks a predefined data model.
- Difficult to organize using traditional databases.

Fun Fact: Approximately 80% of enterprise data is unstructured, but structured data remains the primary fuel for traditional Machine Learning models.

Deep Dive: Structured Data

- **Characteristics:**
  - Fits neatly into rows and columns (tables).
  - Each column has a specific, strict data type (e.g., date, integer, currency).
  - Easily queried using SQL or spreadsheet formulas.
- **Examples:**
  - Relational databases (MySQL, PostgreSQL).
  - Spreadsheets (Excel, Google Sheets).
  - Student grades, inventory logs, bank transaction records.

Deep Dive: Unstructured Data

- **Characteristics:**
  - No predefined format or organization.
  - Requires advanced techniques like NLP (Natural Language Processing) or Computer Vision to extract meaning.
  - Rich in context but computationally expensive to process.
- **Examples:**
  - Emails, social media posts, WhatsApp chat logs.
  - Images, audio files, video recordings.
  - PDF documents, Word files, website text.

Structured vs. Unstructured: A Comparison

Feature	Structured Data	Unstructured Data
Format	Tabular (Rows & Columns)	Free-form text, media
Searchability	Easy (Exact matches)	Hard (Requires semantic search)
Storage	Relational Databases	Data Lakes, File Systems
ML Usage	Traditional ML (Regression, Trees)	Deep Learning, NLP, CV

The Goal of Week 2

We will focus on **Structured Data** because it is the most common starting point for business analytics and foundational Machine Learning projects.

Types of Data in AI

Within structured and unstructured categories, data takes several specific forms:

- **Text:** Words, sentences, documents, reviews.
- **Numbers:** Quantitative measurements, prices, ages.
- **Images:** Pixels, shapes, colors, medical scans.
- **Audio:** Sound waves, speech recordings, music.

In this course, we will primarily interact with **Text** and **Numbers** as they form the basis of the tabular datasets we will explore today.

Data Type: Text

- **What it is:** Any sequence of characters, words, or sentences.
- **Examples:** Customer reviews, email bodies, product descriptions, chat logs.
- **AI Application:** Natural Language Processing (NLP).
  - Sentiment analysis (Is the customer happy or angry?).
  - Chatbots and virtual assistants.
  - Language translation.

Business Context

An AI can read thousands of text reviews to automatically categorize them as "Positive", "Neutral", or "Negative".

Data Type: Numerical

Numerical data represents quantities and can be measured or counted.

Discrete Data

- Countable, whole numbers.
- *Examples:* Number of children, items sold, clicks on a website.

Continuous Data

- Measurable, can include decimals.
- *Examples:* Temperature, salary, distance, time.

ML algorithms use numerical data to calculate averages, trends, and mathematical relationships.

Data Type: Categorical

Categorical data represents characteristics, groups, or labels. Even if they contain numbers, they represent categories.

Nominal Data

- No inherent order.
- *Examples:* Gender, City (Nairobi, Mombasa), Blood Type.

Ordinal Data

- Has a clear, meaningful order.
- *Examples:* Satisfaction (Low, Med, High), Education Level.

Note: ML models cannot process text categories directly; they must be converted into numbers (e.g., Nairobi = 1, Mombasa = 2) before training.

Data Type: Images & Audio

Images

- Composed of pixels (numbers representing color intensity).
- **AI Use:** Computer Vision (Facial recognition, medical X-ray analysis, quality control in manufacturing).

Audio

- Composed of sound waves (frequencies over time).
- **AI Use:** Speech-to-text, voice assistants (Siri, Alexa), music recommendation.

While powerful, image and audio data require specialized Deep Learning models, which we will touch upon conceptually later in the course.

Interactive Discussion

"Identifying data in everyday Kenyan business contexts."

Interactive Discussion: Kenyan Context

Let's apply what we've learned to the local business ecosystem.

Discussion Prompt

Think about the industries that drive the Kenyan economy: Finance, Agriculture, Retail, and Telecommunications.

- Where do we see **Structured Data**?
- Where do we see **Unstructured Data**?

Let's explore three major sectors to see how data manifests in the real world.

Case Study: Mobile Lending Apps

Examples: M-Shwari, Fuliza, Tala, Branch

Structured Data

- Loan amount requested.
- Repayment date.
- M-Pesa transaction IDs.
- Credit score (Numerical).

Unstructured Data

- Customer support chat logs.
- Voice notes from field agents.
- SMS text content (for alternative credit scoring).

AI Application: Structured data predicts repayment probability; unstructured SMS data helps assess financial behavior for users with no formal credit history.

Case Study: Agri-Tech & Supply Chain

Examples: DigiFarm, Twiga Foods

Structured Data

- Crop yield per acre.
- Fertilizer quantity purchased.
- Delivery timestamps.
- Market prices.

Unstructured Data

- Drone imagery of farms.
- Localized weather reports (Text).
- Photos of crop diseases.

AI Application: Computer Vision (Unstructured) detects crop disease from phone photos, while predictive models (Structured) forecast market prices based on historical yield data.

Case Study: Retail & E-Commerce

Examples: Jumia, Local Supermarkets

Structured Data

- Sales receipts / Invoices.
- Inventory stock levels.
- Customer demographic profiles.

Unstructured Data

- Product reviews and ratings text.
- Social media mentions (Twitter/X).
- Product images.

AI Application: Recommendation engines use structured purchase history, while sentiment analysis uses unstructured review text to gauge brand health.

Part 2: Datasets & CSV Files

"Understanding the standard format for ML data."

What is a Dataset?

- A **Dataset** is a structured collection of data, typically presented in a tabular format.
- It is the specific set of information we feed into a Machine Learning model to teach it how to make predictions.
- In business terms, a dataset is like a comprehensive ledger or registry that holds all the historical records relevant to a specific problem.

Example

A "Customer Churn Dataset" contains historical records of thousands of customers, including their demographics, usage patterns, and whether they eventually left the company.

What is a CSV File?

- **CSV** stands for **C**omma-**S**eparated **V**alues.
- It is a plain text file that stores tabular data.
- Each line in the file represents a **row** in the table.
- Commas (or sometimes tabs/semicolons) separate the values in each **column**.
- It is the universal standard for exchanging data between different systems, databases, and ML tools.

Why CSV is the AI Standard

Why not just use Excel (.xlsx)?

- Excel files contain formatting, colors, formulas, and macros.
- CSVs contain **pure data only**.
- CSVs are lightweight, universally compatible across all operating systems, and easily read by Python, R, and cloud AI platforms.

When you export data from a database or an ERP system to train an AI model, it is almost always exported as a CSV.

Anatomy of a Dataset: Headers

- **Headers** are the names given to each column.
- They appear in the very first row of the dataset.
- Headers act as the "labels" or "titles" for the data beneath them.
- In Machine Learning, headers represent the **Features** (the variables we measure) and the **Target** (the variable we want to predict).

Example Headers

```
CustomerID, Age, City, MonthlyCharge, ChurnStatus
```

Anatomy of a Dataset: Columns

- **Columns** run vertically down the dataset.
- Each column represents a specific **Feature** or **Variable**.
- All data within a single column should ideally be of the same **Data Type** (e.g., all numbers, or all text).
- **Target Column:** One specific column is usually designated as the "Target" or "Label" – this is what the AI is trying to predict.

Rule of Thumb

If a column contains mixed data types (e.g., numbers and text mixed together), it will cause errors when training an ML model.

Anatomy of a Dataset: Rows

- **Rows** run horizontally across the dataset.
- Each row represents a single **Record**, **Observation**, or **Data Point**.
- A row contains all the feature values for one specific entity (e.g., one customer, one transaction, one patient).
- The total number of rows determines the "size" of your dataset (e.g., "We trained the model on 10,000 rows of data").

Visualizing the Anatomy

Sample Dataset

CustomerID	Age	City	Churned
101	34	Nairobi	No
102	45	Mombasa	Yes
103	28	Kisumu	No

- **Headers:** CustomerID, Age, City, Churned (4 Columns).
- **Columns:** Features (Age, City) and Target (Churned).
- **Rows:** 3 individual customer records (Observations).

Tool Setup: Excel vs. Google Sheets

We will use spreadsheet software to view and navigate our CSV datasets.

Microsoft Excel

- Powerful offline tool.
- Great for large datasets.
- Advanced formulas and Power Query.

Google Sheets

- Browser-based.
- Excellent for collaboration.
- Cloud storage and easy sharing.

Both tools can easily open, edit, and export CSV files. Ensure you have access to at least one before proceeding to the hands-on task.

Hands-on Mini-Task: Preparation

Step 1: Download the Data

- Download the sample raw CSV file (`sample_customers.csv`) provided by the instructor via the learning portal.
- Save it to your `Downloads` or `Documents` folder.

Step 2: Open the File

- Open Google Sheets (File → Import → Upload) OR Right-click the file and select "Open with Excel".

Hands-on Mini-Task: Execution

Troubleshooting: The "Single Column" Problem

If your data opens as a single, long column of text separated by commas:

- **In Excel:** Go to `Data > Text to Columns`, choose "Delimited", and select "Comma".
- **In Google Sheets:** Select the column, go to `Data > Split text to columns`, and choose "Comma" as the separator.

Your goal is to see the raw text transform into a clean, structured, readable table with distinct columns and rows.

Hands-on Mini-Task: Observation

Now that your dataset is open, identify the following:

1. **Headers:** What are the names of the columns?
2. **Dimensions:** Roughly how many rows (customers) and columns (features) are there?
3. **Data Types:**
   - Which columns contain pure **Numbers**?
   - Which columns contain **Text/Categories**?
   - Are there any **Dates**?

Instructor Check

Raise your hand if you have successfully identified the data types in your sample file.

Day 1 Assignment

Task

- Find **one example** of structured data and **one example** of unstructured data in your daily life or workplace.
- Write a short paragraph (3-4 sentences) explaining how an AI might use *both* types of data together to solve a specific business problem.

Submission

Upload your short paragraph to the Mediacrest learning portal by tomorrow morning (Friday). This will help us contextualize tomorrow's data cleaning session.

Q&A and Wrap-up

Questions?

"Data is the foundation. Without it, AI is just theory."

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